

4. Algorithm Design

Planning Approach

The route planner will use a **graph-based path planning approach**.

The base algorithm used will be:

A* (A-star)

A* is chosen because it provides efficient pathfinding and allows the use of heuristics to guide the search.

Planning Model

The planner will represent the environment using a **grid-based map representation**.

Steps:

1. Define map boundaries
2. Convert obstacles into blocked grid cells
3. Define start node and goal node
4. Execute A* search
5. Generate route nodes
6. Apply route smoothing

Obstacle Processing

Obstacle geometry will be obtained from:

- OpenStreetMap building data
- user-defined restricted zones

These geometries will be converted into blocked regions within the planning grid.

Height Handling

Although the UI is two-dimensional, the planner will incorporate height as a numeric parameter.

Example rule:

if $\text{drone_height} < \text{obstacle_height} + \text{safety_margin}$
path must avoid obstacle

Height values will be used to influence route feasibility.